Variational Self-Distillation for Remote Sensing Scene Classification

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***Abstract*— Supported by deep learning techniques, remote sensing scene classification, a fundamental task in remote image analysis, has recently obtained remarkable progress. However, due to the severe uncertainty and perturbation within an image, it is still a challenging task and remains many unsolved problems. In this article, we note that regular one-hot labels cannot precisely describe remote sensing images, and they fail to provide enough information for supervision and limiting the discriminative feature learning of the network. To solve this problem, we propose a variational self-distillation network (VSDNet), in which the class entanglement information from the prediction vector acts as the supplement to the category information. Then, the exploited information is hierarchically distilled from the deep layers into the shallow parts via a variational knowledge transfer (VKT) module. Notably, the VKT module performs knowledge distillation in a probabilistic way through variational estimation, which enables end-to-end optimization for mutual information and promotes robustness to uncertainty within the image. Extensive experiments on four challenging remote sensing datasets demonstrate that, with a negligible parameter increase, the proposed VSDNet brings a significant performance improvement over different backbone networks and delivers state-of-the-art results.**

***Index Terms*— Class entanglement information, hierarchical knowledge transfer, remote sensing scene classification, self-distillation.**

I. Introduction

REMOTE sensing image analysis, aiming to understand the content of the ground surface, is an active research area in the community of Earth observation [1]–[4].

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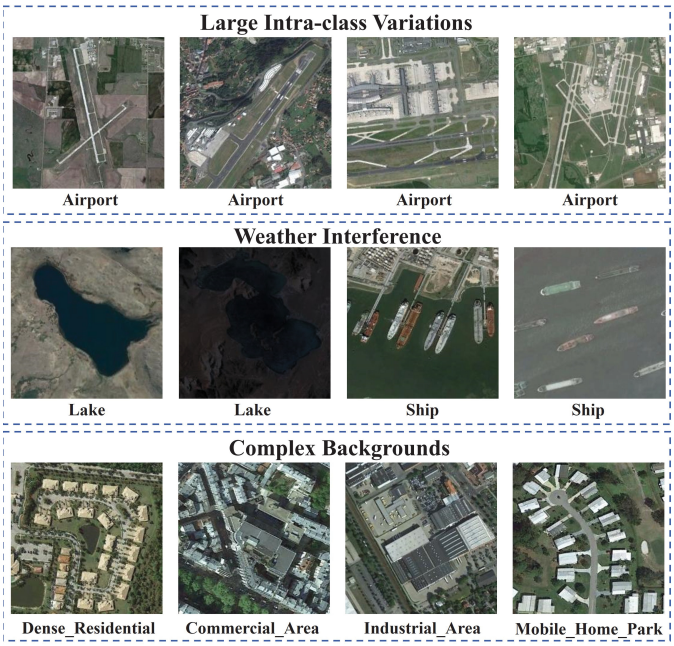


Fig. 1. Severe uncertainty and perturbation exist within remote sensing images. From top to bottom, three groups of images reflect the large intraclass variations, weather interference, and complex backgrounds, respectively.

In particular, remote sensing scene classification, as the fundamental task in remote sensing image analysis, has received considerable attention due to its wide applications in urban management, emergency rescue, and security surveillance. However, remote sensing scene classification remains to be an unsolved problem and its difficulties can be divided into the following three aspects:

1) *Large Intraclass Variations:* Due to variations in shape, scale, and spatial distribution, the appearances of remote sensing images can have large differences even if they belong to the same class. As shown in the first row of Fig. 1, four images are all selected from the "airports". However, they look quite different.
2) *Weather Interference:* Remote sensing images are more easily affected by weather conditions. For example, as shown in the second row of Fig. 1, adverse illumination and visibility conditions introduce severe perturbation into the images, bringing difficulties for accurate classification.

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3) *Complex Backgrounds:* Due to the large visual field, remote sensing images often contain complex backgrounds [5]. As illustrated in the last row of Fig. 1, the backgrounds include roads, trees, lawns, as well as other kinds of objects, and some of them are even more salient than the foreground.

In summary, severe uncertainty and perturbation exist within remote sensing images, increasing the difficulties of discriminative feature learning. More importantly, to handle the uncertainty and perturbation within an image, a more robust feature learning approach is expected.

Recently, with remarkable performance in expressive feature learning, convolutional neural networks (CNNs) have obtained superior performance in remote sensing scene classification. In regular CNNs, to train a specialized model for feature extraction, the one-hot ground-truth label is employed to guide feature learning. The one-hot label assumes an equal distance between the correct category and the others. In other words, it only brings category information (i.e., the input image belongs to which category) but fails to provide interclass relationship, which may hinder sufficient model training. Taking the figures in Fig. 1 as examples, "dense_residential to commercial_area" and "dense_residential to airport" have the same distance in one-hot label. Nevertheless, "dense_residential" is more similar to "commercial_area" than "airport". Under this condition, category information alone cannot precisely describe the images, which leads to inadequate supervision in the training. Therefore, besides the category information, more fine-grained information should be included to supervise the model for discriminative feature learning.

In this article, to address the problem mentioned above, we propose a novel method, named variational self-distillation network (VSDNet), which employs the prediction vector from the network itself as the soft label and transfers its inherent knowledge to improve the feature learning. As depicted in Fig. 2, although the soft label assigns probabilities to incorrect classes, the relative probabilities of the incorrect answers are exceptionally informative about class entanglement [6]–[9]. The hidden information encapsulated in these secondary probabilities is regarded as "dark knowledge", which shows great effectiveness in promoting feature learning. Inspired by this observation, in the training of VSDNet, we exploit the entanglement information from the prediction vector and incorporate it to provide the training instruction besides the one-hot ground-truth label. Moreover, our self-distillation framework enables the knowledge transfer within the single network, avoiding the training of extra teacher models. Specifically, the deeper layers in the network act as the teacher model to guide the previous parts, and the class entanglement information is introduced into the backbone network layer-by-layer, which conforms to the connectivity philosophy of CNNs as well [10], [11].

To promote the robustness of knowledge distillation, we design the variational knowledge transfer (VKT) Module to maximize the mutual information between each student-teacher pair. However, as indicated in prior work [12]–[14], mutual information is computationally challenging. To overcome this problem, in our work, we hypothesize that the teacher's prediction vector can be generated from the student's one and formulate the knowledge transfer in VKT module as a generative process. Thus, we can employ variational estimation to solve the intractable conditional distribution and construct a lower bound on the mutual information, based on which the end-to-end optimization is achieved. In the generation process, we expect the student's prediction vector to be a "good input" based on which the teacher's prediction can be reconstructed. Thus, through the joint training of the backbone network and VKT module, the student branch gradually mimics the teacher part and absorbs its inherent knowledge, which is coincident with the idea of self-distillation. Moreover, through variational estimation, we establish a probabilistic model and the class entanglement information is mapped into a distribution rather than a deterministic vector. As supported by prior works [15], [16], compared to the deterministic representation, the probabilistic distribution is more elastic to fit the randomness within the inputs, which could better handle the uncertainty and idiosyncratic perturbation existing in the remote sensing images. Therefore, VKT module can achieve more robust knowledge transfer.

It is worth mentioning that, in the training, the proposed VSDNet only involves negligible parameter increase. During the inference, all the auxiliary classifiers are discarded and only the final classifier is utilized to generate the prediction. Compared to the inference in the plain network, VSDNet does not introduce any extra parameters or computational overhead.

The main contributions are summarized as follows:

1) We propose VSDNet to adaptively exploit class entanglement information and hierarchically distill it from the deep layers into the shallow parts. In this way, besides the category information, more fine-grained information is provided in the training, contributing to discriminative feature learning.
2) We design a VKT module to maximize the mutual information between each student–teacher pair and promote the robustness of knowledge distillation. In VKT module, we utilize variational estimation to solve the intractable conditional distribution, based on which the end-to-end optimization is enabled. By doing so, a probabilistic model is established, which is more robust to the uncertainty and idiosyncratic perturbation within the remote sensing images.
3) We perform extensive experiments with different backbones on four challenging remote sensing scene classification datasets, where VSDNet works in a plug-and-play manner and produces state-of-the-art results with negligible parameter increase.

II. Related Work

A. Remote Sensing Scene Classification

In the early years, hand-crafted feature descriptors, such as histogram of oriented gradient (HOG) [17], scale-invariant feature transform (SIFT) [18], and bag-of-visual-words (BoVW) [19], were widely utilized in remote sensing scene classification [20]–[23]. However, designing these well-performing hand-crafted features is time-consuming [24], and

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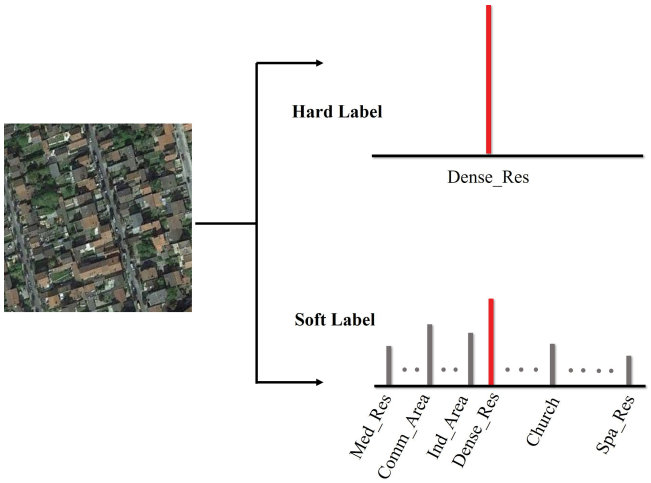


Fig. 2. Differences between hard labels and soft labels. For the hard label, only the ground-truth class is activated. In contrast, soft label assigns values to incorrect classes, which reflects the entanglement among different classes.

the representation ability is limited, which restricts their further improvement for classification accuracy.

Recently, for the purpose of discriminative feature learning, CNNs have become the most promising approach in remote sensing scene classification. With the increased availability of remote sensing data and parallel computing resources, CNN-based methods show great capacity and achieve convincing performance. Specifically, some methods aim to localize the significant regions containing discriminative clues. For example, attention recurrent convolutional network (ARCNet) designs a novel recurrent attention structure to locate significant regions and accelerate the convergence [25]. Key filter bank network (KFBNet) utilizes a key filter bank to capture discriminative local details and retain global information at the same time [26]. Fang *et al.* design a spectral-wise attention mechanism with soft weights applied to selectively emphasize informative spectral features and suppress less useful spectral features [27]. Beyond regular CNNs, Zhang *et al.* [28] employ capsule network to capture the hierarchical structure of the entities in the images and encode the spatial information. Meanwhile, some methods generate discriminative representations by fusing multiple CNN features. Feature aggregation convolutional neural network (FACNN) utilizes a progressive aggregation strategy to fuse the intermediate features according to different spatial structure information [29]. Wang *et al.* extract multiscale multilevel features and aggregate them together at different levels with a two-branch deep feature fusion module [5]. Moreover, inspired by fine-grained visual categorization, the encoding-based method also shows great efficacy in remote sensing scene classification. Wang *et al.* propose a novel multigranularity canonical appearance pooling to learn granular feature representation, where the global representation is generated based on the second-order statistics [30]. Xu *et al.* first map the samples to the manifold space of the lie group and calculate regional covariance to extract lie group features [31]. Then, a novel lie group regional influence network with multidilation pooling module is proposed for final prediction. Furthermore, Hydra utilizes a coarsely optimized network as the Hydra's body and builds several convolutional branches to form multiple Hydra's heads [32]. Then, the final prediction is obtained via the ensemble strategy, which promotes performance with a relatively small training time.

The aforementioned CNN-based methods mainly utilize the one-hot ground-truth label to supervise the network. However, as analyzed in Section I, one-hot label only provides the category information and results in the inadequate supervision for remote sensing scene classification. In our VSDNet, motivated by knowledge distillation, we exploit the class entanglement information to promote training, leading to a more powerful model for discriminative feature learning.

B. Knowledge Distillation

Knowledge distillation is originally proposed by Hinton *et al.* for model compression [33], in which the teacher's prediction vector is employed as the soft label to train the student. By doing so, the knowledge stored in the soft label, indicating the degree of class entanglement, is transferred to the student model, which not only contributes to generating a compact model but also promotes the feature learning. Since then, fruitful knowledge distillation methods have been proposed. In deep mutual learning (DML), unlike the distillation between a well-trained teacher model and a student model, an ensemble of students learn collaboratively and teach each other through joint training [34]. Moreover, FitNet employs not only the prediction vector but also the intermediate representations to improve the student model [35]. To our best knowledge, this is the first work to distill the intermediate feature maps between the teacher and student. Inspired by this, AT presents a knowledge distillation loss based on the summed p-norm of convolutional feature activations along the channel dimension [36]. Furthermore, Walawalkar *et al.* design a novel knowledge distillation framework with multiple student ensembles, in which each model learns unique representations from the data distribution, and better performance is obtained via the ensemble mechanism [37].

Generally, in the aforementioned works, teacher models and student models work in their own ways. Namely, knowledge distillation is performed between different networks, which means extra training overhead for the multiple models. Recently, to save computational resources, self-distillation has been proposed and received increasing research attention, in which the student and teacher models share the same network. For example, Zhang *et al.* divide the network into several sections, and then the knowledge stored in the deeper layers is distilled into the shallow ones [8], [38]. Additionally, self-attention distillation (SAD) exploits attention maps extracted from deeper layers as the distillation targets for its shallow layers, and it reports impressive performance in lane detection without the need of additional labels and external supervision [39].

Inspired by these works, self-distillation has been introduced into remote sensing image analysis to improve feature learning. Multimodal CNN with per-source Self-Distillation (MMCNN_SD) utilizes a multibranch network for different input sources, and the final ensemble classification results are

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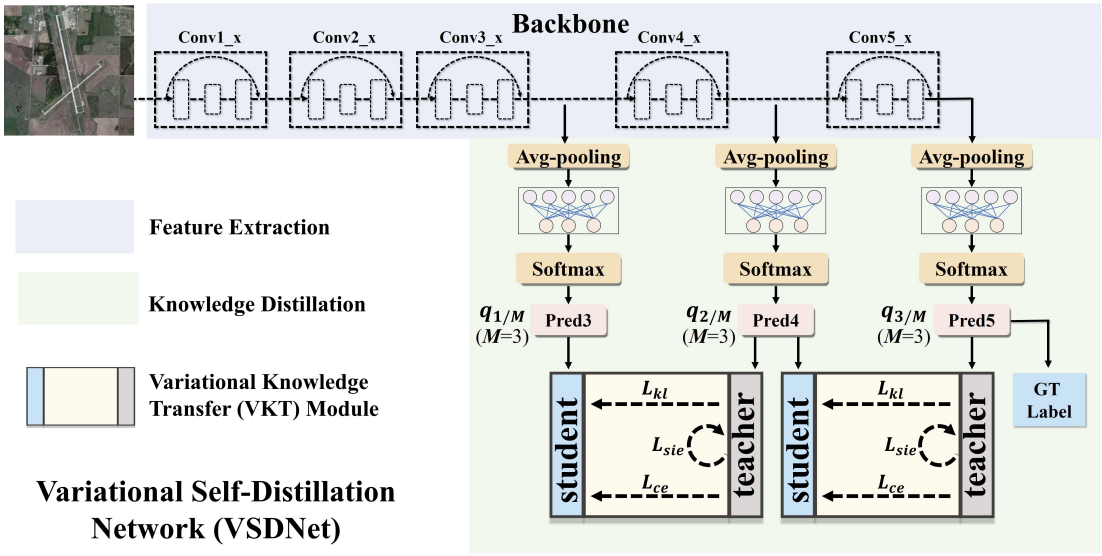


Fig. 3. Details of our VSDNet equipped on the ResNet backbone. The backbone network consists of five convolutional blocks. In addition to the classifier located at the end of the backbone, we attach two auxiliary classifiers at intermediate layers. As shown, the three classifiers form two "student-teacher" pairs. Each pair is fed into a VKT module to transfer the class entanglement information. Notably, two intermediate classifiers with VKT modules will be discarded in the test, and Pred5 is utilized as the final prediction.

distilled to supervise each branch [40]. Duan *et al.* design a self context distillation module (SCDM), in which the feature maps from deeper layers are employed to supervise the features from earlier layers [41]. Chai *et al.* design a novel bidirection distillation approach composed of backward boundaries distillation and forward inner distillation, which contribute to obtaining detailed boundaries and reliable cloud detection results, respectively, [42]. In contrast to these works, we establish a probabilistic model instead of the deterministic one to achieve knowledge transfer, in which the class entanglement information is mapped into a distribution. In this way, the proposed VSDNet is more robust to uncertainty and idiosyncratic perturbation within the remote sensing images. Furthermore, our VSDNet does not rely on extra convolution branches, and the distillation process is achieved within a single path network.

III. Method

A. Preliminary

Given a classification network $E(\cdot) \sim \theta$ and an input image $\boldsymbol{x}$ drawn from K classes, we denote the ground-truth label as a one-hot vector $\boldsymbol{y} = [0, 0, 1, \ldots, 0]^T \in R^K$. As shown in Fig. 3, the input image is processed by several convolutional blocks, which produces multiple intermediate feature maps. Then, based on the final classifier at the end of the backbone, we can obtain the probability vector $\boldsymbol{q}$ through a softmax function as

$$q^k = \frac{\exp(z^k)}{\sum_{k=1}^{K} \exp(z^k)} \tag{1}$$

where k represents the kth class among all the categories, $\boldsymbol{z}$ is the logits produced by the fully connected layer and q^k is the probability value of kth class in the prediction vector $\boldsymbol{q}$.

In prior self-distillation works [8], [38], multiple auxiliary classifiers are attached at the intermediate layers. Each classifier is denoted as $f_{m/M}, m = \{1, 2, \ldots, M\}$, and their corresponding predictions are denoted as $\boldsymbol{q}_{m/M}$, where M indicates the total number of classifiers and m is the index of each classifier (from shallow to deep). Then, the deep portion of the network is utilized as the teacher model to squeeze the knowledge into shallow layers. To this end, the cross-entropy loss is widely employed as follows:

$$L_{\mathrm{SD}} = -\frac{1}{N} \sum_{i=1}^{N} \hat{\boldsymbol{q}}^T_{m_t/M,i} \log\left(\hat{\boldsymbol{q}}^S_{m_s/M,i}\right) \tag{2}$$

in which $\hat{\boldsymbol{q}}$ is the prediction normalized by the hyperparameter (i.e., temperature T_{em}) and N is the batch size. m_t and m_s are the indices of prediction vectors from teacher and student, respectively. Generally, $m_t > m_s$. In this way, the distillation is organized as a deterministic model, where class entanglement information is transferred from teacher branch to student one and then flows into the backbone network. However, due to the severe uncertainty and perturbation within remote sensing images, calculating the class entanglement via a deterministic feature representation is vulnerable. Thus, it is critical to exploit entanglement information and distill its inherent knowledge in a more robust way.

B. Variational Knowledge Transfer

To achieve more robust knowledge transfer, we design a VKT module to maximize the mutual information between each student-teacher pair. The details of the VKT module are illustrated in Fig. 4. In this part, we first present the mutual information with its derived variational lower bound. Then, Monte Carlo sampling based on the reparameterization trick [43] is introduced.

1) Mutual Information: Mutual information is a fundamental quantity for measuring the relationship between random variables [12]. From the perspective of information theory, knowledge transfer can be expressed as retaining high mutual information between the layers of the teacher and student

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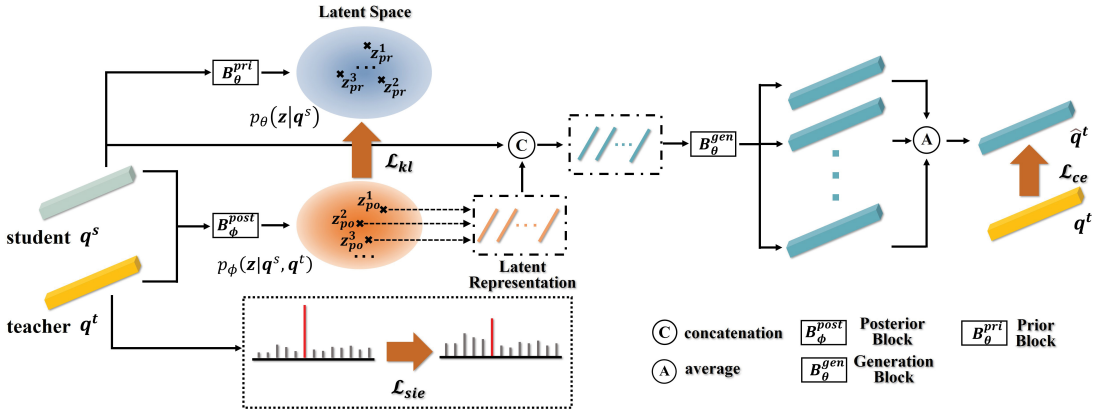


Fig. 4. Structure of VKT module. The prediction vectors from the student and teacher branches, $\boldsymbol{q}^s$ and $\boldsymbol{q}^t$, are fed into the VKT as the inputs. B_ϕ^{post} and B_θ^{pri} calculate the estimated posterior and prior distribution, respectively. Based on the sampled latent representation with the student's prediction vector, B_θ^{gen} produces the predicted $\hat{\boldsymbol{q}}^t$. Within VKT module, $\mathcal{L}_{\text{sie}}$, $\mathcal{L}_{\text{kl}}$, and $\mathcal{L}_{\text{ce}}$ are calculated to ensure the sufficient knowledge transfer.

parts [44]. Here, given a student-teacher pair with their prediction vectors $\boldsymbol{q}^s$ and $\boldsymbol{q}^t$, respectively, our goal is to maximize the mutual information between them, enabling the sufficient knowledge transfer between two branches. At first, the mutual information can be formulated as

$$\begin{aligned} I(\boldsymbol{q}^t;\boldsymbol{q}^s) &= H(\boldsymbol{q}^t) - H(\boldsymbol{q}^t|\boldsymbol{q}^s) \\ &= -E_{\boldsymbol{q}^t}\left[\log(\boldsymbol{q}^t)\right] + E_{\boldsymbol{q}^t,\boldsymbol{q}^s}\left[\log p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s)\right] \end{aligned} \tag{3}$$

where the first term is the entropy of probability vector $\boldsymbol{q}^t$, which can be explicitly calculated as

$$-\mathcal{L}_{\text{sie}} = -\frac{1}{N}\sum_{i=1}^{N} \boldsymbol{q}_i^t \log \boldsymbol{q}_i^t. \tag{4}$$

Nevertheless, the second term attempts to find the conditional distribution $p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s)$ for a student-teacher pair, which is intractable and computational challenging. From the view of distillation, $\boldsymbol{q}^t$, the prediction from the teacher branch, is the expected output of $\boldsymbol{q}^s$. Thus, in our VKT module, we hypothesize that the teacher's prediction vector could be reconstructed from the student's one and formulate the knowledge transfer as a generative process. Then, the variation estimation technique is utilized to solve the intractable conditional distribution. To be more specific, according to [43], we derive the variational lower bound of $\log p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s)$ as

$$\begin{aligned} \log p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s) \geq & -\text{KL}\left(p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t)||p_\theta(z|\boldsymbol{q}^s)\right) \\ & + E_{p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t)}\left[\log p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s,z)\right] \end{aligned} \tag{5}$$

where KL$(\cdot)$ denotes the Kullback-Leibler (KL) divergence between the posterior distribution $p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t)$ and the prior distribution $p_\theta(z|\boldsymbol{q}^s)$. Specifically, taking the prediction vectors $\boldsymbol{q}^s$ and $\boldsymbol{q}^t$ as inputs, the posterior block B_ϕ^{post} is utilized to calculate a mean vector $\boldsymbol{\mu}_{\text{po}}$ and a variance vector $\boldsymbol{\sigma}_{\text{po}}^2$ for the distribution $p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t)$

$$p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t) = \mathcal{N}\left(z;\boldsymbol{\mu}_{\text{po}},\boldsymbol{\sigma}_{\text{po}}^2\right). \tag{6}$$

Similarly, the prior block B_θ^{pri} is employed to calculate the distribution $p_\theta(z|\boldsymbol{q}^s)$ as

$$p_\theta(z|\boldsymbol{q}^s) = \mathcal{N}\left(z;\boldsymbol{\mu}_{\text{pr}},\boldsymbol{\sigma}_{\text{pr}}^2\right). \tag{7}$$

Via (6) and (7), we estimate the distribution of class entanglement information and generate the latent representation z. Then, to maximize the first term in (5), the KL loss is calculated as follows:

$$\mathcal{L}_{\text{kl}} = \frac{1}{N}\sum_{i=1}^{N} p_\phi(z_i|\boldsymbol{q}_i^s,\boldsymbol{q}_i^t)\log\left(\frac{p_\phi(z_i|\boldsymbol{q}_i^s,\boldsymbol{q}_i^t)}{p_\theta(z_i|\boldsymbol{q}_i^s)}\right). \tag{8}$$

Meanwhile, the second term in (5) measures the difference between the prediction generated from the $\boldsymbol{q}^s$ and its expected output $\boldsymbol{q}^t$. Specifically, to maximize $E_{p_\phi(z|\boldsymbol{q}^s,\boldsymbol{q}^t)}[\log p_\theta(\boldsymbol{q}^t|\boldsymbol{q}^s,z)]$ for the generated prediction $\hat{\boldsymbol{q}}^t$, we define the loss $\mathcal{L}_{\text{ce}}$ as follows:

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N} \boldsymbol{q}_i^t \log \hat{\boldsymbol{q}}_i^t. \tag{9}$$

Then, according to prior works [15], [45], we can incorporate $E_{\boldsymbol{q}^t,\boldsymbol{q}^s}[\log \text{p}(\boldsymbol{q}^t|\boldsymbol{q}^s)]$ as a data-driven term into the optimization objective. Thus, based on (4) and (5), we attain the empirical objective for end-to-end optimization of (3)

$$\mathcal{L}_{\text{MI}}^{s,t} \approx \mathcal{L}_{\text{sie}} + \mathcal{L}_{\text{kl}} + \mathcal{L}_{\text{ce}}. \tag{10}$$

In the aforementioned generation process, to promote the accurate reconstruction from $\boldsymbol{q}^s$ to $\boldsymbol{q}^t$, the student branch is expected to produce a prediction vector containing homogeneous information with the teacher's one. This motivation conforms to the idea of distillation and enables the knowledge transfer from deep layers to shallow parts [8], [38].

Meanwhile, minimizing $\mathcal{L}_{\text{sie}}$ increases the amount of information in the prediction vector, which contributes to generating the informative soft label. Experiments demonstrate that this is more effective than softening the prediction vectors via the hyperparameter. On the other hand, minimizing $\mathcal{L}_{\text{sie}}$ also reduces the confidence of the network for its prediction. As advocated in [46], [47], preventing the classifier from being too confident in its prediction is beneficial for improving the generalization ability, especially for data with the large variation and uncertainty.

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Algorithm 1 VKT

Learning: Input N pairs of student-teacher prediction vectors $\{\boldsymbol{q}_i^s, \boldsymbol{q}_i^t\}_{i=1}^N$; Initialized parameters θ, ϕ.
repeat:

$$\text{Posterior:} \begin{cases} \boldsymbol{\mu}_{po}, \boldsymbol{\sigma}_{po} \leftarrow B_\phi^{post}(\boldsymbol{q}_i^t, \boldsymbol{q}_i^s) \\ \boldsymbol{z}_{po} \leftarrow \boldsymbol{\mu}_{po}(\boldsymbol{q}_i^s) + \boldsymbol{\epsilon} * \boldsymbol{\sigma}_{po}(\boldsymbol{q}_i^s), \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}) \end{cases}$$

$$\text{Prior:} \begin{cases} \boldsymbol{\mu}_{pr}, \boldsymbol{\sigma}_{pr} \leftarrow B_\theta^{pri}(\boldsymbol{q}_i^s) \\ \boldsymbol{z}_{pr} \leftarrow \boldsymbol{\mu}_{pr}(\boldsymbol{q}_i^s) + \boldsymbol{\epsilon} * \boldsymbol{\sigma}_{pr}(\boldsymbol{q}_i^s), \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}) \end{cases}$$

Generation: $\hat{\boldsymbol{q}}^t \leftarrow B_\theta^{gen}(\boldsymbol{q}_i^s, \boldsymbol{z}_{po})$
$\boldsymbol{g} \leftarrow \nabla_{\theta,\phi}(\mathcal{L}(\theta, \phi, \boldsymbol{q}_i^t, \boldsymbol{q}_i^s, \boldsymbol{\epsilon}) + \mathcal{L}_{GT})$
$\theta, \phi \leftarrow$ Update parameters using gradients $\boldsymbol{g}$.
until convergence
return: θ, ϕ

2) Monte Carlo Sampling: To solve the nondifferentiable problem in the sampling process, we leverage the reparameterization trick [43] to generate the latent representation, $\boldsymbol{z}^{(l)} = \boldsymbol{\mu} + \boldsymbol{\epsilon}^{(l)} \odot \boldsymbol{\sigma}$, where $\odot$ indicates the element-wise multiplication and $\boldsymbol{\epsilon}^{(l)}$ is sampled from an auxiliary noise distribution $\mathcal{N}(\mathbf{0}, \boldsymbol{I})$. Moreover, we totally draw L samples $\{\boldsymbol{z}_{\text{po}}^{(l)}\}_{l=1}^L$ from the posterior distribution $p_\phi(\boldsymbol{z}|\boldsymbol{q}^s, \boldsymbol{q}^t)$. Each sampled latent representation $\boldsymbol{z}_{\text{po}}^{(l)}$ is concatenated with $\boldsymbol{q}^s$ and then fed into the generation block B_θ^{gen} to produce the reconstructed teacher's prediction vector $\hat{\boldsymbol{q}}^t$. Taking the average of L samples, we obtain the final predicted $\hat{\boldsymbol{q}}^t$ as

$$\hat{\boldsymbol{q}}^t = \frac{1}{L}\sum_{l=1}^{L} p_\theta\left(\boldsymbol{q}^t|\boldsymbol{q}^s, \boldsymbol{z}_{\text{po}}^{(l)}\right), \quad \boldsymbol{z}_{\text{po}}^{(l)} \sim \boldsymbol{p}_\phi(\boldsymbol{z}|\boldsymbol{q}^s, \boldsymbol{q}^t). \tag{11}$$

In this way, we fully utilize the distribution to produce the similarity factor, establishing a probabilistic model instead of a deterministic one. As advocated in [15], [16], the probabilistic model is more robust to the uncertainty and idiosyncratic perturbation within remote sensing images. Thereby, the more robust knowledge transfer is achieved. Experiments demonstrate that three samples are enough to estimate the class entanglement information for a given image, which is a good trade-off between computational overhead and accuracy. We summarize the learning procedure of the VKT module in Algorithm 1.

C. Implementation Within VSDNet

As shown in Fig. 3, taking ResNet-50 as an example, the backbone network is consists of five convolutional blocks, each of which produces a feature map F in $N \times C \times W \times H$. N, C, W and H represent the batch size, channel, width and height of the feature map F, respectively. Besides the final classifier at the end of the backbone, we attach two auxiliary classifiers after conv3_x and conv4_x (this setting is proven to be optimal via the ablation study). Namely, there are three classifiers in VSDNet, and thus M equals to 3. Each classifier, from shallow to deep, produces the prediction vectors, which are denoted as pred3, pred4, and pred5, respectively. On the one hand, the final prediction, pred5, is supervised by the one-hot ground-truth label, which calculates the $\mathcal{L}_{\text{GT}}$ and provides the category information to guide the network training. On the other hand, as depicted in Fig. 3, we establish two student-teacher pairs, i.e., pred3-pred4 and pred4-pred5. Each pair is fed into the VKT module to maximize the mutual information and enable the knowledge distillation. In this way, the class entanglement information is transferred from the teacher branch to student one, which guides the feature learning within the backbone network. Notably, B_ϕ^{post}, B_θ^{pri} and B_θ^{gen} can be organized in any form, such as a fully connected block or a convolutional block. Here, we utilize two fully connected layers to produce the $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$, respectively. Overall, the total loss of VSDNet is denoted as

$$\mathcal{L}_{\text{total}} = \sum_{i}^{N}\left(\mathcal{L}_{\text{MI}}^{3,4} + \mathcal{L}_{\text{MI}}^{4,5} + \mathcal{L}_{\text{GT}}\right) \tag{12}$$

where the two numbers in the superscript of $\mathcal{L}_{\text{MI}}$ denote the indices of the involved student-teacher pair.

In the test, all the auxiliary classifiers with the following VKT module will be discarded, and only the final classifier at the end of the backbone is employed to generate the prediction. Namely, we only utilize the auxiliary classifiers with VKT module to promote the feature learning in the training. In the test, the well-trained backbone network has the capacity to extract discriminative feature solely, which does not increase extra parameters. More importantly, experiments demonstrate our method works in a plug-and-play manner that could improve the performance over various backbones.

IV. Experiments

To evaluate the effectiveness of the proposed VSDNet, we perform extensive experiments with various backbones on four public and challenging remote sensing datasets, including NWPU-RESISC45 [48], aerial image dataset (AID) [49], OPTIMAL-31 [25] and UC-Merced [50]. The experimental results demonstrate that our VSDNet is widely applicable and brings performance improvement over various backbone networks.

A. Dataset

In this section, we provide a brief introduction to the four datasets utilized in the experiments. Meanwhile, some selected examples are shown in Fig. 5. It is worthwhile to mention that, all the training ratios for four datasets are employed according to previous works [20], [25], [51], [52].

1) NWPU-RESISC45 [48]: NWPU-RESISC45 is one of the largest datasets in remote sensing scene classification with 31 500 images from 45 different scene categories in total. The spatial resolution of most images varies from 30 to 0.2 m/pixel. According to prior works [32], [51], [53], we perform two groups of experiments for NWPU-RESISC45 with training ratios of 10% and 20%, and the remaining 90% and 80% of the data are utilized for testing.

2) AID [49]: AID is a large-scale remote sensing scene classification dataset, containing 30 scene categories with 10 000 samples in total. The pixel resolution of each image ranges from 8 to 0.5 m. When conducting the experiments, two widely used training ratios are 20% and 50%, and the rest 80% and 50% are employed for testing [20].

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Fig. 5. We show some examples from four remote sensing scene classification datasets. From top to bottom are NWPU-RESISC45, AID, OPTIMAL-31 and UC-Merced. The category information is noted below each image.

3) OPTIMAL-31 [25]: OPTIMAL-31 has 31 categories in total and each category contains 60 images. According to [25], we utilize 80% of the samples to train the model and the remaining 20% are employed in the evaluation.

4) UC-Merced [50]: The UC-Merced dataset consists of 21 different scene classes, each of which contains 100 images, and the spatial resolution is 0.3 m/pixel. Following the split strategy in [5], [31], [52], we conduct two groups of experiments with training ratios of 50% and 80%, and the rest samples are utilized to evaluate the performance.

B. Implementation Details

We utilize stochastic gradient descent to optimize the model, in which the initial learning rate is set to 0.01 and the momentum is set to 0.9. In the training stage, we first resize the short edge of the image to 448 with the ratio unchanged. Then, we randomly crop a patch of size 448×448 from the resized image. Afterward, random horizontal flip and random rotation are employed to generate the training samples. In the test, we first resize the image via the same strategy as in the training and then center-crop a 448×448 patch to obtain the inputs. To reduce the influence of randomness and produce more robust result, we repeat the main experiments five times and report the average result.

In the experiments, we utilize the overall accuracy to evaluate the performance. The overall accuracy is formulated as the number of correct predictions divided by the total number of test samples. Moreover, we also draw the confusion matrix to analyze the classification errors across different classes. It is important to mention that we only note accuracy values that are at least 0.01. Thus, in the illustrated confusion matrix, the sums of some rows do not equal to 1.

C. Network Architecture

To verify the compatibility of the proposed method, we test the performance of VSDNet with VGG-16 [54], ResNet-34 [55], and DenseNet-121 [56]. The experimental results demonstrate that our method achieves a consistent performance improvement over different backbones.

In our VSDNet, besides the final classifier placed at the end of the backbone, we add auxiliary classifiers at different intermediate layers for each specific network. In VGG-16, we attach auxiliary classifiers at the relu3_3 and relu4_3 layers. In ResNet-34, we mount auxiliary classifiers after conv3_x and conv4_x. As for DenseNet-121, two auxiliary classifiers are placed at the end of dense_block_2 and dense_block_3. Based on these auxiliary classifiers, we maximize the mutual information between different prediction vectors and transfer the inherent knowledge to the backbone network gradually.

D. Performance and Comparison

To evaluate the performance of the proposed method, we compare VSDNet with various scene classification approaches. The results are listed in Tables I–IV. Each table is split into five parts. From the top to bottom, the first three parts report the accuracy of the models based on VGG-16, ResNet-34, and DenseNet-121, respectively. The fourth part reports the accuracy of the methods equipped on other backbones (e.g., AlexNet, Inception-v3) or established on its own structure. The fifth part shows the performance of our method. The best performance in each table is noted in bold. Generally speaking, VSDNet brings comprehensive performance improvements over different backbones and achieves state-of-the-art results on four challenging datasets. The detailed analysis for each dataset is elaborated below.

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TABLE I

ACCURACY COMPARISON WITH DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET

Method	Training ratio	
	10%	20%
VGG-16 [54]	88.17±0.32	91.62±0.33
HW-CNN-VGG16 [53]	-	94.38±0.16
KFBNet-VGG16 [26]	90.27±0.02	92.54±0.03
CapsNet-VGG16 [28]	85.08±0.13	89.18±0.14
MG-CAP-VGG16 [30]	90.83±0.12	92.95±0.13
ResNet-34 [55]	90.24±0.21	92.78±0.18
DenseNet-121 [56]	91.36±0.22	94.03±0.18
KFBNet-DenseNet121 [26]	93.08±0.14	95.11±0.10
CapsNet-Inception-v3 [28]	89.03±0.21	92.60±0.11
LGRIN [31]	91.91±0.15	94.43±0.16
Hydra [32]	92.44±0.34	94.51±0.21
Triplet Network [57]	-	92.33±0.20
VSDNet-VGG16	90.12±0.24	93.35±0.19
VSDNet-ResNet34	92.13±0.16	94.68±0.13
VSDNet-DenseNet-121	**93.24±0.11**	**95.67±0.11**

TABLE II

ACCURACY COMPARISON WITH DIFFERENT METHODS ON THE AID

Method	Training ratio	
	20%	50%
VGG-16 [54]	92.63±0.23	94.94±0.19
HW-CNN-VGG16 [53]	-	96.98±0.33
GBNet-VGG16 [52]	92.20±0.23	95.48±0.12
MG-CAP-VGG16 [30]	93.34±0.18	96.12±0.12
CapsNet-VGG16 [28]	91.63±0.19	94.74±0.17
FACNN-VGG16 [29]	-	95.45±0.11
KFBNet-VGG16 [26]	94.27±0.02	97.19±0.07
ResNet-34 [55]	93.68±0.21	95.01±0.23
EFPNet-ResNet34 [5]	94.02±0.21	94.50±0.30
DenseNet-121 [56]	94.80±0.29	95.92±0.28
KFBNet-DenseNet121 [26]	95.50±0.27	97.40±0.10
LGRIN [31]	94.74±0.23	97.65±0.25
CapsNet-Inception-v3 [28]	93.79±0.13	96.32±0.12
VSDNet-VGG16	95.11±0.21	97.16±0.19
VSDNet-ResNet34	96.00±0.18	97.28±0.14
VSDNet-DenseNet-121	**96.73±0.15**	**97.95±0.10**

TABLE III

ACCURACY COMPARISON WITH DIFFERENT METHODS ON THE OPTIMAL-31

Method	Training Ratio: 80%
VGG-16 [54]	92.50±0.41
ARCNet-VGG16 [25]	92.70±0.35
GBNet-VGG16 [52]	93.28±0.27
KFBNet-VGG16 [26]	95.12±0.13
ResNet-34 [55]	93.91±0.37
DenseNet-121 [56]	94.87±0.32
KFBNet-DenseNet121 [26]	95.60±0.63
ARCNet-AlexNet [25]	85.75±0.35
VSDNet-VGG16	94.91±0.26
VSDNet-ResNet34	96.22±0.21
VSDNet-DenseNet-121	**96.95±0.17**

TABLE IV

ACCURACY COMPARISON WITH DIFFERENT METHODS ON THE UC MERCED

Method	Training ratio	
	50%	80%
VGG-16 [54]	96.18±0.38	98.11±0.32
ARCNet-VGG16 [25]	96.81±0.14	99.12±0.40
GBNet-VGG16 [52]	97.05±0.19	98.57±0.48
FACNN-VGG16 [29]	-	98.81±0.24
CapsNet-VGG16 [28]	95.33±0.18	98.81±0.22
MG-CAP-VGG16 [30]	-	99.00±0.10
KFBNet-VGG16 [26]	-	99.76±0.24
ResNet-34 [55]	96.84±0.25	98.28±0.26
EFPNet-ResNet34 [5]	96.19±0.13	99.14±0.22
DenseNet-121 [56]	97.62±0.26	98.73±0.20
KFBNet-DenseNet121 [26]	-	**99.88±0.12**
CapsNet-Inception-v3 [28]	97.59±0.16	99.05±0.24
LGRIN [31]	98.61±0.22	98.97±0.31
VSDNet-VGG16	97.91±0.22	99.55±0.19
VSDNet-ResNet34	98.49±0.18	99.67±0.18
VSDNet-DenseNet-121	**98.85±0.13**	99.86±0.11

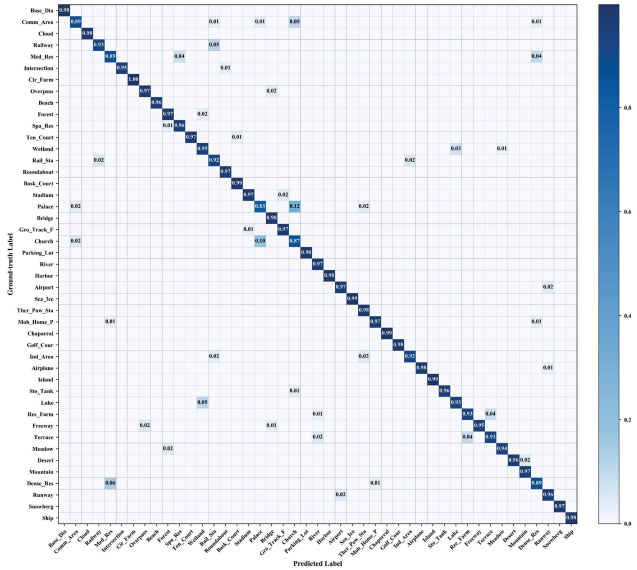


Fig. 6. Confusion matrix of VSDNet based on DenseNet-121 over NWPU-RESISC45 under the 20% training ratio.

1) Results on NWPU-RESISC45: As shown in Table I, VSDNet achieves the best performance with the accuracy of 93.24% ± 0.11 and 95.67% ± 0.11 under the 10% and 20% training ratios. Specifically, with 10% of the data for training, our VSDNet gains 1.95%, 1.89%, and 1.88% improvements when equipped on VGG-16, ResNet-34, and DenseNet-121, respectively. Meanwhile, 1.73%, 1.90%, and 1.64% improvements are obtained when the model is trained with 20% samples. Moreover, in Fig. 6, we illustrate the confusion matrix of VSDNet based on DenseNet-121 under the training ration of 20%. As shown, the proposed VSDNet produces an accuracy over 95% on 33 different categories.

2) Results on AID: As listed in Table II, for two different training ratios, VSDNet reports the highest accuracy with DenseNet-121 utilized as the backbone network. Specifically,when equipped on VGG-16, ResNet-34, and DenseNet-121, our method achieves 2.48%, 2.32%, and 1.93% improvements under 20% training ratio, and 2.22%, 2.27%,

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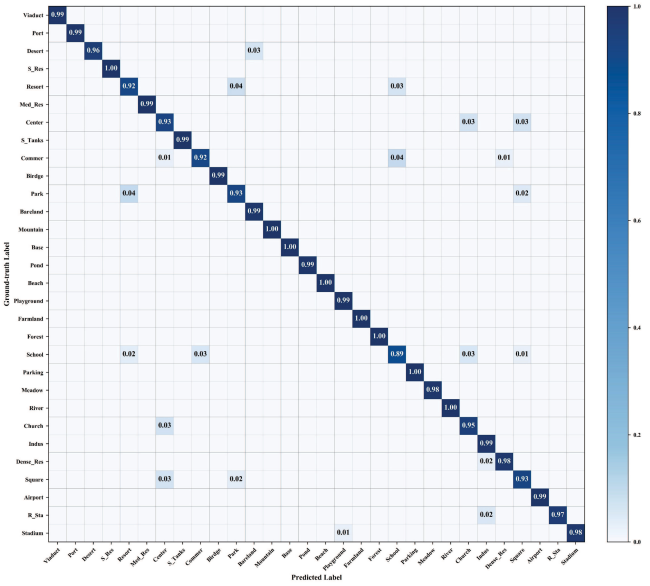


Fig. 7. Confusion matrix of VSDNet based on DenseNet-121 over AID under the 50% training ratio.

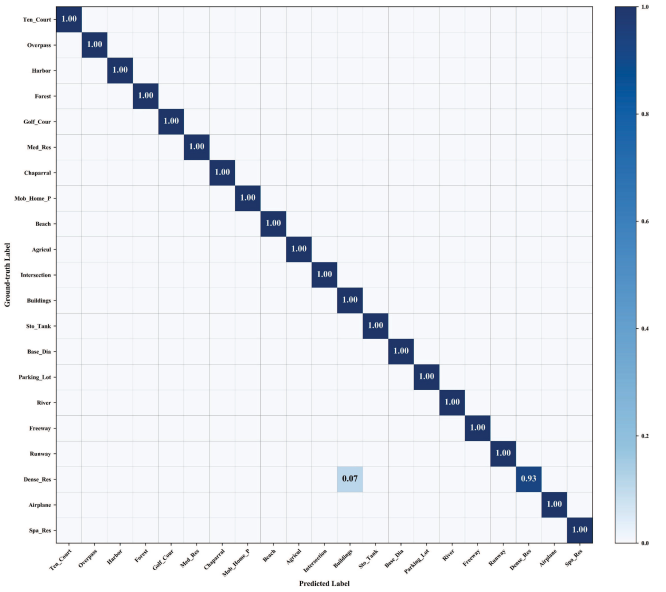


Fig. 9. Confusion matrix of VSDNet based on DenseNet-121 over UC-Merced under the 80% training ratio.

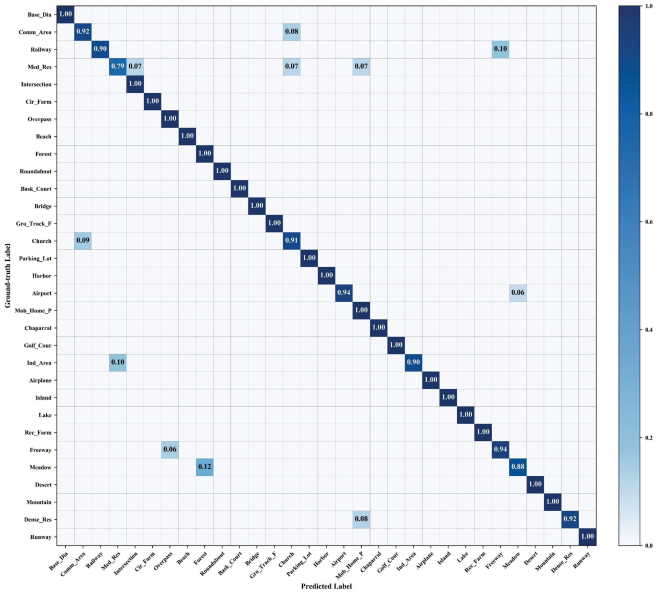


Fig. 8. Confusion matrix of VSDNet based on DenseNet-121 over OPTIMAL-31 under the 80% training ratio.

and 2.03% accuracy increases are achieved under 50% ratio. The confusion matrix is shown in Fig. 7, and we can find that 21 of the 30 categories achieve a classification accuracy over 98%.

3) Results on OPTIMAL-31: As listed in Table III, we utilize 80% of the samples to train our VSDNet and obtain the best performance based on the DenseNet-121. Meanwhile, our method improves VGG-16, ResNet-34, and DenseNet-121 by 2.41%, 2.31%, and 2.08%, respectively. We further illustrate the confusion matrix in Fig. 8, from which we can see that 22 of the 31 categories obtain an accuracy of 100%

4) Results on UC-Merced: The accuracy on UC-Merced is reported in Table IV. Similar to the results on other datasets, VSDNet equipped on DenseNet-121 delivers competitive accuracy. Also, our method improves the performance significantly compared to three different backbone networks, VGG-16, ResNet-34, and DenseNet-121. On the one hand, 1.73%, 1.65%, and 1.23% improvements are obtained under the training ratio of 50%. On the other hand, 1.44%, 1.39%, and 1.13% accuracy increases are achieved as well under the ratio of 80%. We draw the confusion matrix in Fig. 9. Almost all the categories obtain a classification accuracy of 100%.

Overall, compared to the previous methods, the proposed VSDNet achieves the superior performance. Specifically, as listed in Tables I, II and IV, compared to [28], VSDNet obtains 5.04%, 3.48%, and 2.58% improvements on NWPU-RESISC45, AID, and UC Merced when both equipped with VGG-16 as the backbone network (in terms of training ratio of 10%, 20%, and 50%, respectively). Meanwhile, although lie group regional influence network (LGRIN) [31] relies on Lie Group Machine Learning as the preprocess to calculate the lie group regional covariance in manifold space, its accuracy are much lower than VSDNet in most conditions.

It is worthwhile to mention that, in the test, all the auxiliary classifiers with VKT modules are removed and only final classifier is utilized to produce the prediction result. In other words, during the test, the procedure of VSDNet is the same with the plain network, such as VGG-16 or ResNet-50. Therefore, compared to the plain network, VSDNet brings universal performance gains without parameter increases or extra computational overhead, which further demonstrates the superiority of our method.

E. Ablation Study

In this section, we carry out the ablation study to investigate the efficacy of the proposed VSDNet. Unless otherwise specified, all the experiments in the ablation study are performed with ResNet-34 on the NWPU-RESISC45 and AID under the training ratio of 10% and 20%, respectively.

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TABLE V

ACCURACY COMPARISON OF VSDNET WHEN THE AUXILIARY CLASSIFIERS ARE ATTACHED AT DIFFERENT POSITIONS

	Configuration	NWPU	AID
1	conv4_x	91.24±0.18	94.96±0.21
2	conv3_x	91.07±0.20	94.82±0.23
3	conv2_x + conv3_x + conv4_x	91.69±0.18	95.51±0.19
4	conv2_x + conv3_x	91.31±0.22	95.20±0.18
5	conv2_x + conv4_x	91.58±0.20	95.38±0.21
6	conv3_x + conv4_x	**92.13±0.16**	**96.00±0.18**

TABLE VI

ACCURACY COMPARISON OF VSDNET WHEN WE UTILIZE DIFFERENT "STUDENT-TEACHER" PAIRS

Student-Teacher Pair	NWPU	AID
Pred3-Pred5, Pred4-Pred5	90.48±0.22	94.11±0.25
Pred3-Pred4, Pred4-Pred5	**92.13±0.16**	**96.00±0.18**

TABLE VII

ACCURACY COMPARISON OF VSDNET WITH RELATED SELF-DISTILLATION WORKS TOWARD CLASSIFICATION

Method	NWPU	AID
BYOT [8]	91.78±0.17	95.46±0.23
HSD [9]	91.93±0.16	95.71±0.19
VSDNet	**92.13±0.16**	**96.00±0.18**

TABLE VIII

ACCURACY COMPARISON OF DIFFERENT WAYS TO GENERATE SOFT LABEL

Method	NWPU	AID
$T_{em}=3$	91.28±0.22	94.93±0.27
$T_{em}=5$	90.81±0.18	94.26±0.24
$\mathcal{L}_{sie}$	**92.13±0.16**	**96.00±0.18**

1) Ablation on Hierarchical Structure: The number and locations of auxiliary classifiers are two important settings in VSDNet. To figure out their influence, we establish different configurations and evaluate the classification accuracy. Specifically, we keep the hierarchical supervision mechanism unchanged, in which each auxiliary classifier is supervised by its nearest deeper classifier. The results are reported in Table V. As shown, the best performance for ResNet-34 backbone is achieved when two auxiliary classifiers are attached at conv3_x and conv4_x. On NWPU-RESISC45, 0.89% and 1.06% decreases are observed (the first row and second row) when we discard the classifiers at conv3_x and conv4_x, respectively. Also, accuracy degrades from 92.13% to 91.69% when we add another classifier at conv2_x. We think this is caused by the coarse information within the shallow layers. Additionally, when we change the positions of two auxiliary classifiers, performance degradation also emerges (the fourth row and fifth row). Similar fluctuations are observed on AID as well.

Additionally, we change the combination of "student-teacher" pairs. Specifically, we utilize Pred5 as the soft label to supervise Pred3 and Pred4 simultaneously. Under this condition, the hierarchical supervision mechanism is broken and the two auxiliary classifiers have the same learning target. As listed in Table VI, the accuracy degenerates from 92.13% to 90.48% on NWPU-RESISC45, while on AID, the accuracy degenerates from 96.00% to 94.11%. According to prior works, different intermediate layers conform to different learning targets [9]. Therefore, supervising two auxiliary classifiers via the same label violates the connectivity and feature hierarchy established in the CNNs [10], [11], resulting in the inferior performance in Table VI.

Overall, these experimental results demonstrate that for ResNet-34, the best performance is achieved when auxiliary classifiers are attached at conv3_x and conv4_x. According to this protocol, we establish the proposed VSDNet on VGG-16 and DenseNet-121.

2) Comparison With Deterministic Distillation: To further demonstrate the efficacy of variational estimation for knowledge distillation, we compare VSDNet with two deterministic self-distillation approaches [8], [9]. The results are listed in Table VII, which reflects the superiority of our probabilistic model. To be more specific, through variational estimation, we map the class entanglement information into the distribution, which is more robust to the uncertainty and idiosyncratic perturbation within the remote sensing images [15], [16]. Therefore, the better performance is achieved.

3) Analysis of $\mathcal{L}_{\text{sie}}$: In regular knowledge distillation, to obtain the informative soft label, a widely employed way is to normalize the logits via the hyperparameter, temperature T_{em}, which is calculated as follows:

$$\hat{q}^k = \frac{\exp\left(z^k / T_{\text{em}}\right)}{\sum_{k=1}^{K} \exp\left(z^k / T_{\text{em}}\right)}. \tag{13}$$

In our VSDNet, instead of the hyperparameter, the self-information entropy loss $\mathcal{L}_{\text{sie}}$ is utilized to produce the soft label. On the one hand, minimizing $\mathcal{L}_{\text{sie}}$ increases the amount of information in the prediction vector, which suppresses the large values and promotes the small ones. This encourages the generation of the soft label. On the other hand, as advocated in [46], [47], increasing the amount of information in the prediction vector reduces the confidence of the network for its prediction, which makes it more robust to uncertainty within the images.

We perform experiments to support this view. At first, we plot the histogram of the maximum value in $\boldsymbol{q}_{M/M}$ (i.e., Pred5) over all training samples. Specifically, $\boldsymbol{q}_{M/M}$ is utilized as the label to supervise $\boldsymbol{q}_{M-1/M}$ (i.e., Pred4). As illustrated in Fig. 10, when equipped with $\mathcal{L}_{\text{sie}}$, the maximum value of most training samples is much smaller than that without $\mathcal{L}_{\text{sie}}$, reflecting the efficacy of $\mathcal{L}_{\text{sie}}$ in generating the soft label. Meanwhile, we replace $\mathcal{L}_{\text{sie}}$ by (13) with different temperatures T_{em}, and compare their results in Table VIII. As shown, softening label via $\mathcal{L}_{\text{sie}}$ achieves the best performance, demonstrating the superiority of our method.

4) Visualization of Soft Label: To better demonstrate the efficacy of the proposed method, we visualize the prediction vector generated by the last classifier (e.g., $f_{M/M}$) in VSDNet.

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TABLE IX
ACCURACY COMPARISON OF DIFFERENT SAMPLE NUMBERS L IN VARIATIONAL ESTIMATION

	The Number of Samples					
	1	2	3	4	5	10
NWPU	91.76±0.24	91.93±0.18	**92.13±0.16**	92.10±0.18	92.09±0.16	92.11±0.10
AID	95.31±0.24	95.76±0.22	**96.00±0.18**	95.96±0.16	95.96±0.18	95.91±0.20

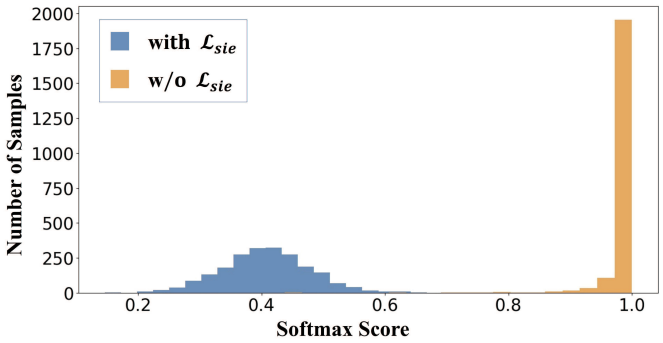


Fig. 10. Histogram of the maximum value in the prediction vector ($\boldsymbol{q}_{M/M}$) when VSDNet equipped with or without $\mathcal{L}_{\text{sie}}$.

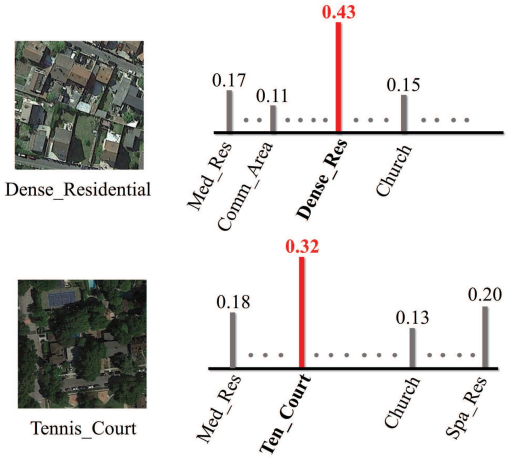


Fig. 11. Visualization of the prediction vector generated by the last classifier (e.g., $f_{M/M}$) of VSDNet. The red column indicates the correct category of the input image. Notably, we only note the responses that are larger than 0.1.

As depicted in Fig. 11, besides the correct category, some response values are also assigned to incorrect categories, which reflects the class entanglement information of the input image. Thus, the prediction vector $q_{M/M}$ can be utilized as the soft label. Taking the first image in Fig. 11 as an example, it belongs to the "Dense_Residential" and there is a response value of 0.43 in the prediction vector. Meanwhile, this image is quite similar to the "Medium_Residential", "Commercial_Area" and "Church". In the prediction vector, these categories are also being activated, showing the entanglement condition of this image. We attribute this phenomenon to the efficacy of L_{sie}, which suppresses the large activations in the prediction vector and promotes the small ones. In this way, L_{sie} increases the amount of information stored in $q_{M/M}$ and encourages the generation of the soft label. Therefore, when we employ $q_{M/M}$ to supervise $f_{M-1/M}$ in the training, class entanglement information is introduced into the network, which could assist the category information in optimizing the network and obtain the better performance.

5) Analysis of Monte Carlo Sampling: In this part, we investigate the effect when different samples are extracted from the distribution $p_\phi(\boldsymbol{z}|\boldsymbol{q}^s, \boldsymbol{q}^t)$. The results are shown in Table IX. The best performance is achieved when L equals to 3. When L becomes larger, the accuracy does not improve. Thus, we believe $L = 3$ is a good trade-off between accuracy and efficiency. It is worth mentioning that the variational estimation is only performed in the training. During the test, only the plain network is utilized to produce final prediction.

V. CONCLUSION

In this article, we propose a novel VSDNet to promote the discriminative feature learning for remote sensing scene classification. We note that the regular one-hot ground-truth label only provides the category information and cannot precisely describe the remote sensing images. To solve this problem, in VSDNet, the prediction vectors from deeper layers are utilized as soft labels to supervise shallower layers, introducing the class entanglement information as the supplement of one-hot label. Meanwhile, to ensure the sufficient knowledge distillation within the network, we maximize the mutual information between different branches via the VKT module. Specifically, the VKT module performs knowledge distillation in a probabilistic way through variational estimation, which enables the end-to-end optimization for mutual information and promotes the robustness to uncertainty within the image. Extensive experiments on four challenging datasets demonstrate that the proposed VSDNet obtains substantial performance improvements over various backbones and reports state-of-the-art classification result.

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